Multi-Omic Analysis: Overlap Analyses

Antoine Chaillon

2026-04-11

Table of contents

Proteomic **3**
3.1 Defined Pathways of Interest . . . 3
3.2 HIV-Associated Inflammatory Signature at Baseline (Naive Groups) . . . 4
3.3 Cannabis Effects on the Proteome in HIV+ Donors (table 2) . . . 5
Overlap Analysis: HIV Suppression vs. Cannabis Rescue . . . 7
Scatter Plot . . . 7
3.4 Cannabis Effects on the Proteome in HIV- Donors . . . 8
Top cannabis-by-HIV interaction proteins in plasma proteomics. (Interaction Table) 9

Transcriptomic **10**
3.6/3.7 Cannabis Effects on Macrophage Transcriptomics in HIV+ Donors . . . 10
3.10 Multi-Omic Convergence . . . 13

> **Note**
>
> Proteomic data from Adam Fields' Laboratory, University of California San Diego (UCSD).

Proteomic

3.1 Defined Pathways of Interest

Copy CSV Excel

Search:

Gene Count	Gene List	Pathway
23	CRP, SAA1, SAA2, FGA, FGB, FGG, ORM1, ORM2, APCS, HP, HPR, LBP, ITIH1, ITIH2, ITIH3, ITIH4, PTX3, SERPINA1, SERPINA3, A2M, TTR, RBP4, CFH	Acute Phase Response
21	CD14, CD68, MRC1, MSR1, ITGAM, ITGAX, CD163, CD80, CD86, TNF, IL1B, IL6, IL8, CXCL10, CCL2, CCL5, HMOX1, MIF, SPP1, TREM2, CSF1R	Macrophage Activation
22	GFAP, S100B, CXCL10, CCL2, TNF, IL1B, IL6, MMP9, TIMP1, VCAM1, ICAM1, VWF, VEGFA, NOS3, NGF, BDNF, CXCL12, CX3CL1, P2RX7, NLRP3, CASP1, IL18	Neuroinflammation
10	NLRP3, CASP1, IL1B, IL18, GSDMD, PYCARD, P2RX7, TXNIP, NEK7, HMGB1	NLRP3 Inflammasome
16	TNF, TNFRSF1A, TNFRSF1B, TRADD, RIPK1, FADD, CASP8, NFKB1, RELA, BIRC2, BIRC3, TRAF2, TRAF3, MAP3K7, IKBKB, IKBKG	TNF Signaling
16	IL6, IL6R, IL6ST, JAK1, JAK2, STAT3, SOCS1, SOCS3, CRP, FGA, FGB, FGG, HP, ORM1, SAA1, SAA2	IL6 JAK STAT3
20	MX1, MX2, OAS1, OAS2, OAS3, OASL, IFIT1, IFIT2, IFIT3, IFITM1, IFITM3, ISG15, ISG20, RSAD2, HERC5, TRIM25, DDX58, IFIH1, CXCL10, IRF7	IFN alpha beta
22	C1QA, C1QB, C1QC, C2, C3, C4A, C4B, C5, CFB, CFD, CFH, CFI, C1R, C1S, MBL2, MASP1, MASP2, CD55, CD59, SERPING1, VTN, CLU	Complement
22	F2, F5, F7, F8, F9, F10, F13A1, F13B, FGA, FGB, FGG, VWF, PROS1, PROC, THBD, PLAT, PLAU, PLG, SERPINE1, SERPINC1, A2M, TFPI	Coagulation Fibrinolysis
19	ATP5PB, ATP5MC1, ATP5F1A, ATP5F1B, NDUFA1, NDUFA2, NDUFB10, NDUFS1, NDUFS2, UQCRC1, UQCRC2, COX4I1, COX5A, COX5B, SDHB, SDHC, SDHD, CYC1, CYCS	Oxidative Phosphorylation

Showing 1 to 10 of 18 entries

Previous 1 2 Next

3.2 HIV-Associated Inflammatory Signature at Baseline (Naive Groups)

Report_Name	NES	pval	padj
Acute Innate Immunity	0.9357	0.5372	0.6155
Acute Phase Response	-2.2894	0	0.0001
Chemokine Signaling	1.7364	0.0128	0.0513
Coagulation Fibrinolysis	1.2555	0.197	0.4727
Complement	-1.7891	0.0066	0.0394
ECM Remodeling	1.0348	0.3966	0.5948
IFN alpha beta	-1.1403	0.2938	0.5875
Lipoprotein Metabolism	0.9107	0.5642	0.6155
Neuroinflammation	1.3189	0.1533	0.4598
Platelet Activation	0.6284	0.9205	0.9205

The significant suppression ($NES = -2.26$, $p = 0.0001$) observed here suggests that HIV+ naive donors exhibit a "cold" or exhausted proteomic baseline.

3.3 Cannabis Effects on the Proteome in HIV+ Donors (table 2)

Copy CSV Excel PDF Search:

Report_Name	HIV+ Daily	HIV+ Moderate	HIV- Daily	HIV- Moderate
Acute Phase Response	**-0.8 (0.73)**	+1.37 (0.11)	-1.95 (0)	-1.22 (0.209)
Neuroinflammation	**-1.86 (0.004)**	-1.5 (0.068)	-0.91 (0.578)	-1.14 (0.305)
TNF Signaling	**+1.15 (0.296)**	+0.96 (0.524)	+0.89 (0.606)	-1.56 (0.031)
IFN alpha beta	**-0.45 (0.997)**	+1.12 (0.324)	-1.22 (0.219)	-0.72 (0.852)
Complement	**-0.73 (0.814)**	-0.82 (0.674)	-1.52 (0.046)	-1.17 (0.259)
Coagulation Fibrinolysis	**-1.32 (0.152)**	-0.87 (0.616)	+0.6 (0.951)	-0.64 (0.924)
Lipoprotein Metabolism	**-1.64 (0.027)**	-1.06 (0.401)	-0.67 (0.895)	+1.27 (0.17)
Vascular Integrity	**-1.53 (0.044)**	-1.04 (0.435)	-0.67 (0.886)	+1.45 (0.062)
ECM Remodeling	**-1.28 (0.18)**	-1.03 (0.423)	+0.69 (0.868)	-0.88 (0.671)
Chemokine Signaling	**-1.67 (0.021)**	-1.49 (0.063)	-1.33 (0.13)	-0.88 (0.657)

Showing 1 to 10 of 12 entries Previous 1 2 Next

The leading edge proteins in Table 3b—likely including Complement C3, Haptoglobin, and Serpins—are significantly lower than in HIV- controls, providing the essential context for why the cannabis-associated "reversal" seen in Table 2 is so striking.

Copy CSV Excel

Search:

Protein	log2FC	t-stat	FDR
CRP	-1.0626	-1.7827	0.9999
SAA2	-0.8154	-2.5135	0.9999
ITIH4	-0.313	-1.7161	0.9999
ITIH3	-0.3073	-1.8492	0.9999
LBP	-0.2645	-0.9662	0.9999
SERPINA1	-0.2407	-1.5121	0.9999
ITIH2	-0.2337	-1.0495	0.9999
FGG	-0.2327	-0.7828	0.9999
APCS	-0.2302	-1.5723	0.9999
CFH	-0.1717	-1.2367	0.9999

Showing 1 to 10 of 11 entries

Previous 1 2 Next

Overlap Analysis: HIV Suppression vs. Cannabis Rescue

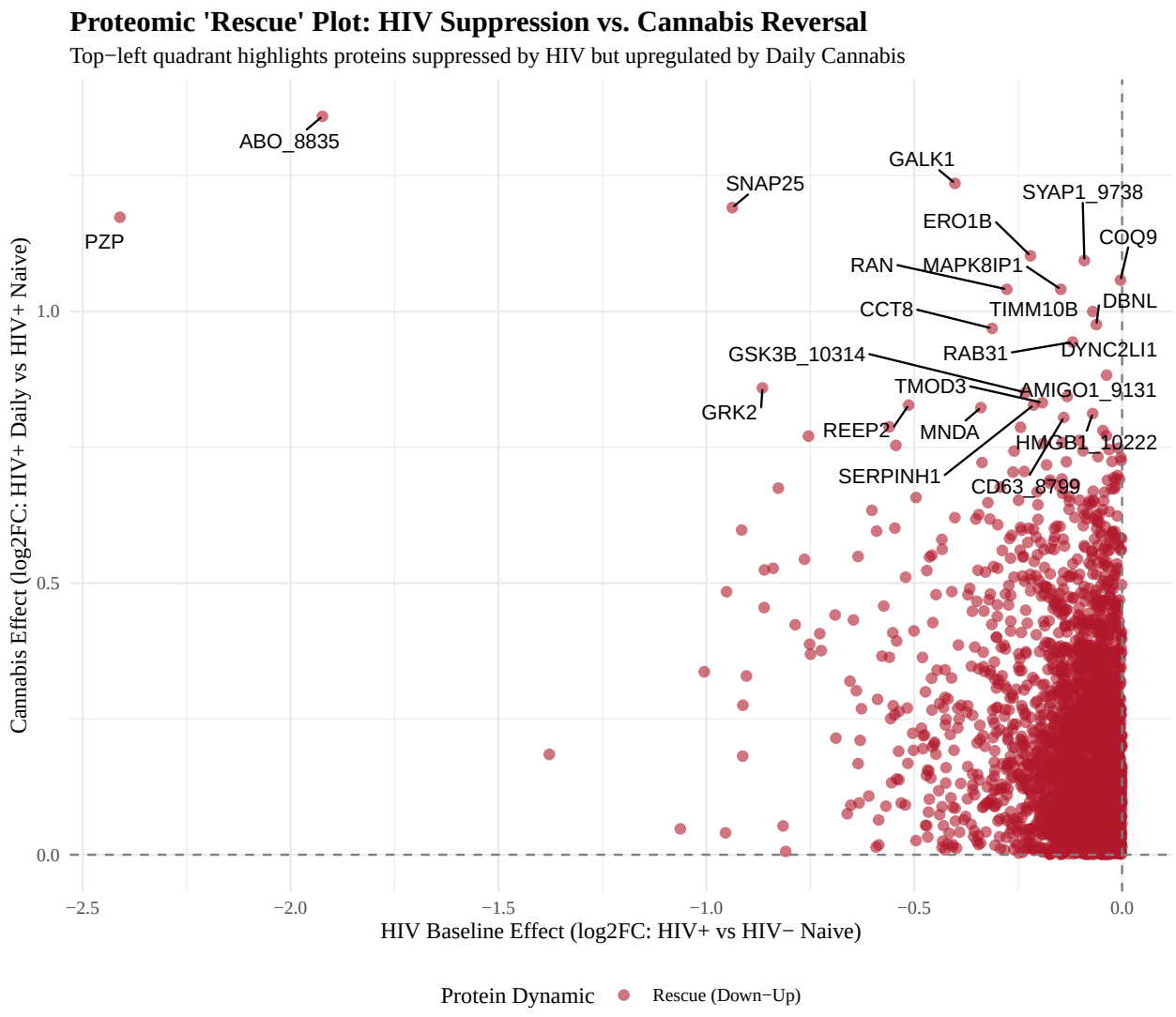
Copy CSV Excel Search:

SYMBOL	t_Baseline	logFC_Baseline	FDR_Baseline	logFC_Cannabis	p_Cannabis
ABO_8835	-1.8634	-1.9234	0.9999	1.3591	0.2165
GALK1	-0.7294	-0.402	0.9999	1.2357	0.0371
SNAP25	-1.0983	-0.9376	0.9999	1.191	0.1907
PZP	-2.781	-2.4104	0.9999	1.1733	0.204
ERO1B	-0.4461	-0.2203	0.9999	1.1024	0.0379
SYAP1_9738	-0.2527	-0.0907	0.9999	1.0936	0.0051
COQ9	-0.0061	-0.0039	0.9999	1.0576	0.1192
MAPK8IP1	-0.3841	-0.1475	0.9999	1.0407	0.0123
RAN	-0.6056	-0.2772	0.9999	1.0407	0.0346
TIMM10B	-0.2111	-0.0706	0.9999	0.9998	0.006

Showing 1 to 10 of 20 entries Previous 1 2 Next

Scatter Plot

To visualize the global remodeling effect, we compared the baseline proteomic signature of HIV infection against the changes associated with daily cannabis use. The resulting four-quadrant plot identifies a distinct population of "Rescue Proteins" (top-left quadrant).



3.4 Cannabis Effects on the Proteome in HIV- Donors

In HIV-negative donors, moderate cannabis use was associated with a notable suppression of TREM2 expression (Mean log_2FC = -0.484, Best p = 0.015), a key regulator of macrophage inflammatory responses.

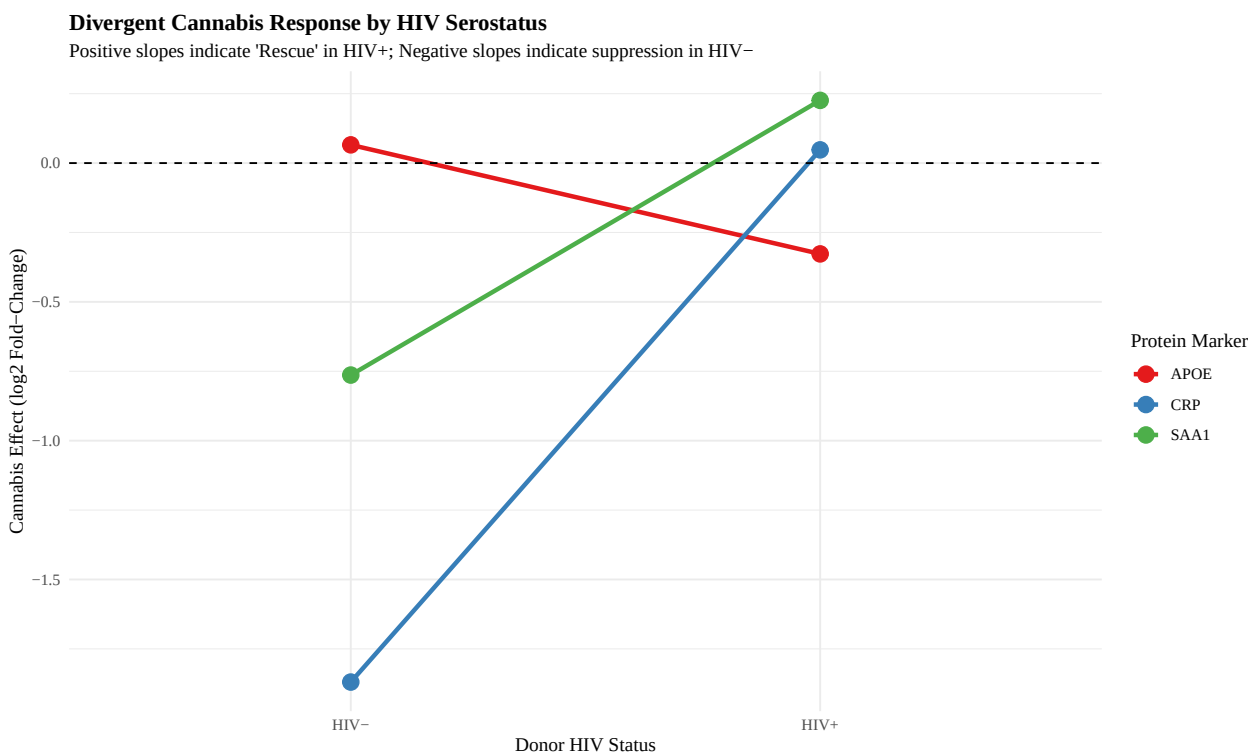
> **Note**
>
> Data Aggregation Note: For proteins represented by multiple Somascan aptamers (e.g., SAA1, S100A9), values are reported as the mean log fold-change across all probes mapping to the specific gene symbol. Statistical significance (p-value) is reported as the minimum value (highest significance) among the associated aptamers to reflect the most robust evidence for the observed interaction.

Top cannabis-by-HIV interaction proteins in plasma proteomics. (Interaction Table)

Copy CSV Excel Search:

Protein	HIVp_logFC	HIVn_logFC	Interaction_Delta	p_val
APOE	-0.3272	0.0656	**-0.3928**	0.0333
CRP	0.0476	-1.8696	**1.9172**	0.0318
CXCL8	-0.931	-0.0217	**-0.9093**	0.0792
FGG	0.1368	-0.0293	**0.1661**	0.1759
SAA1	0.2259	-0.7636	**0.9895**	0.03
SAA2	0.053	-0.912	**0.965**	0.0466

Showing 1 to 6 of 6 entries Previous 1 Next



Results Summary:Formal interaction testing (*Table* 3) revealed that for several major inflammatory markers, the response to cannabis was inverted by HIV status. Specifically, acute-phase reactants SAA1, SAA2, and CRP were suppressed by cannabis in HIV-negative donors but significantly elevated or "restored" in HIV-positive donors. These divergent slopes (Interaction $p < 0.05$) highlight that the immunomodulatory effects of cannabinoids are context-dependent and fundamentally modified by the underlying viral inflammatory state.

Transcriptomic

3.6/3.7 Cannabis Effects on Macrophage Transcriptomics in HIV+ Donors

> **Note**
>
> Vehicle only

Copy CSV Excel Search:

Pathway	HIV+ Moderate	HIV+ Daily	HIV- Moderate	HIV- Daily
IFN-a/b	-0.82 (0.8541)	-1.52 (0.0097)	1.64 (0.0041)	-1.27 (0.0896)
Oxidative Phosphorylation	-1.83 (0.0033)	-1.81 (<0.001)	-1.1 (0.1987)	-1.48 (0.0027)
Complement System	0.76 (0.9141)	-1 (0.4549)	1.68 (<0.001)	-1.66 (<0.001)
TNFa Signaling	-2.56 (0.0034)	-2.34 (<0.001)	1.92 (<0.001)	-1.15 (0.1562)
NLRP3 Inflammasome	-1.6 (0.0468)	-1.16 (0.2933)	1.36 (0.1260)	-1.2 (0.2518)
Neuroinflammation	-1.99 (0.0072)	-1.26 (0.1826)	1.55 (0.0397)	-1.62 (0.0249)
Macrophage Activation	-2.32 (<0.001)	-2.03 (<0.001)	2.04 (<0.001)	0.82 (0.6962)
Endocannabinoid	0.79 (0.7585)	-0.99 (0.4775)	1.04 (0.4203)	-1 (0.4623)

Showing 1 to 8 of 8 entries Previous 1 Next

In HIV+ macrophages, cannabis use was associated with a sweeping suppression of inflammatory and metabolic pathways compared to cannabis-naive donors.

- **Moderate Cannabis Users**: GSEA revealed a robust downregulation of **TNFalpha Signaling** (NES = -2.57, p = 0.003), Macrophage Activation (NES = -2.31, p = 0.001$), and Neuroinflammation (NES = -1.99, p = 0.005).
- **Targeted Suppression**: Notably, the **NLRP3 Inflammasome** axis—a critical driver of chronic HIV-associated inflammation—was significantly suppressed (NES = -1.57, p = 0.045).

Table 1: Top Interaction Genes: Statistical proof of divergent cannabis effects.

Copy CSV Excel Search:

Gene	Interaction LFC	p-value	Effect
CXCL3	-5.3679	0	HIV-Dependent Suppression
IL1B	-3.8013	0.0032	HIV-Dependent Suppression
TNF	-5.5688	0.0048	HIV-Dependent Suppression
CD14	-1.0665	0.3439	HIV-Dependent Suppression
TNF	-3.2168	0.5735	HIV-Dependent Suppression
TNF	-0.8611	0.9605	HIV-Dependent Suppression

Showing 1 to 6 of 6 entries Previous 1 Next

In contrast, the transcriptomic signature in HIV- macrophages showed a paradoxical shift toward activation.

- **Divergent Response:** Among HIV- moderate users, we observed significant **positive** enrichment in **Macrophage Activation** (NES = +2.04, p < 0.001), **TNFalpha Signaling** ($NES = +1.92, p < 0.001$), and **Neuroinflammation** ($NES = +1.55, p = 0.040$).
- **Host-Environment Interaction:** This divergence, where the same exposure (cannabis) acts as a suppressor in HIV+ cells but a mild activator in HIV- cell, provides definitive evidence that the biological effect of cannabis is fundamentally dependent on the underlying immune landscape of the host.

To visualize this, we compare the response to an inflammatory stimulus (IL1B) across groups.

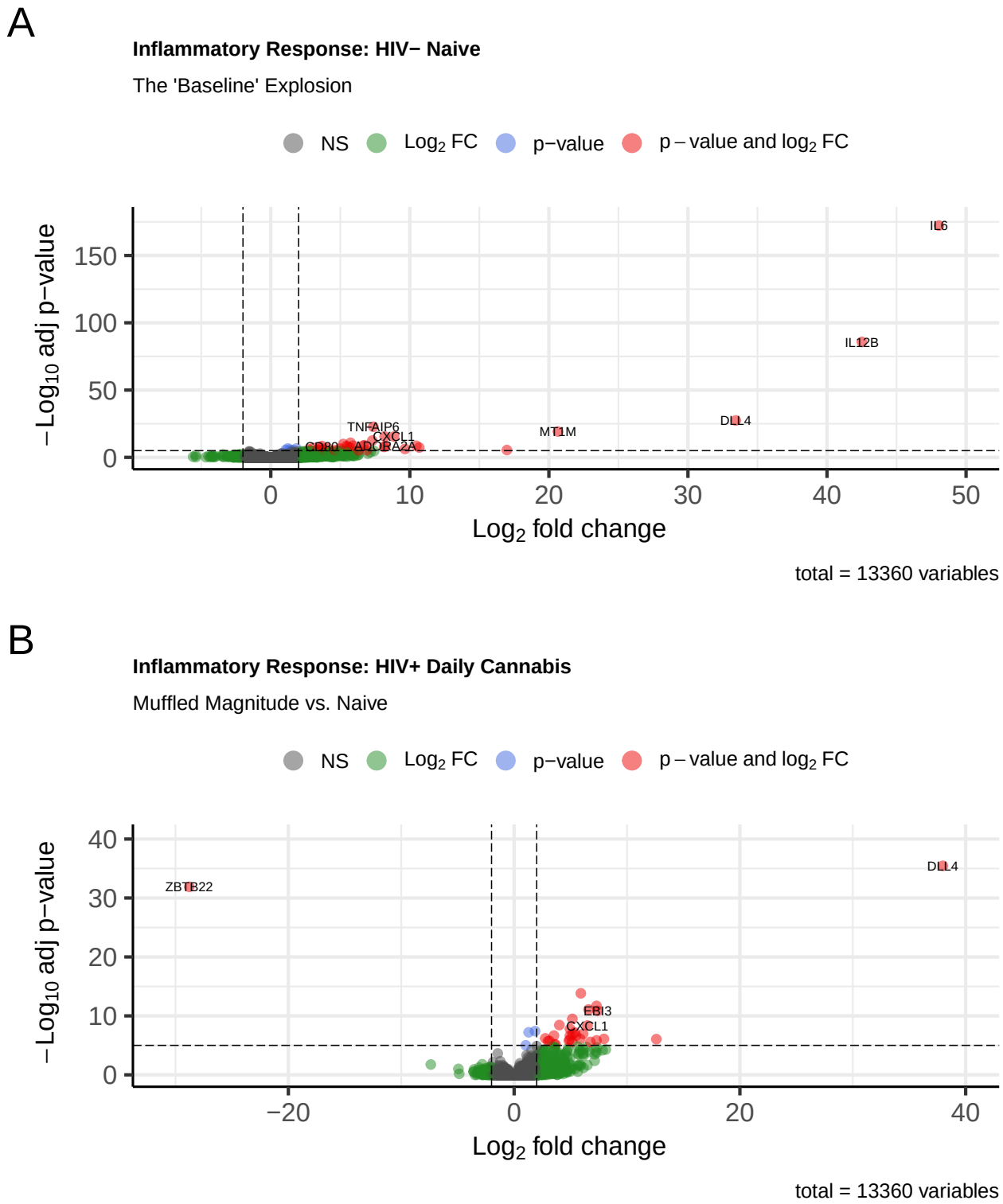


Figure 1: Divergent Inflammatory Potential. (A) Baseline explosion in HIV-naive cells. (B) Muffled response in HIV+ daily users.

(A) The Baseline Inflammatory Explosion: In HIV-negative, cannabis-naive macrophages, stimulation with IL-1beta induces a massive transcriptomic response, characterized by the high-

magnitude induction of core inflammatory drivers including *TNF*, *IL1B*, and *CXCL3* (Log2FC > 4.0, $p < 10^{-10}$).

(B) The Muffled Response in HIV+ Daily Users: In contrast, macrophages from HIV-positive donors with daily cannabis exposure exhibit a significantly "muffled" response to the same IL-1β stimulus. Key inflammatory mediators show a dramatic reduction in both fold-change magnitude and statistical significance (e.g., *TNF* Interaction $Log2FC = -5.57, p = 0.004$).

3.10 Multi-Omic Convergence

Copy CSV Excel Search:

Feature	NES_mRNA	p_mRNA	NES_Prot	p_Prot	p_Interaction	Direction in HIV+
Acute Innate Immunity	1.1298	0.2895	-0.6852	0.8619	0.9057	↑ Enriched
Acute Phase Response	0.8873	0.5725	1.3948	0.0918	0.4603	↑ Enriched
Chemokine Signaling	2.6123	0	-1.508	0.0401	0	↑ Enriched
Coagulation Fibrinolysis	1.5167	0.0616	-0.8828	0.6335	0.317	↑ Enriched
Complement	2.4368	0	-0.8374	0.6908	0.0165	↑ Enriched
ECM Remodeling	1.3124	0.1467	-1.0392	0.424	0.6764	↑ Enriched
IFN-α/β Pathway	1.6648	0.0224	1.1329	0.2974	0.169	↑ Enriched
IL-6/JAK/STAT3 Axis	1.3887	0.11			0.001	↑ Enriched
Lipoprotein Metabolism	0.9107	0.545	-1.0873	0.3524	0.0046	↑ Enriched
Macrophage Activation	2.6838	0			0	↑ Enriched

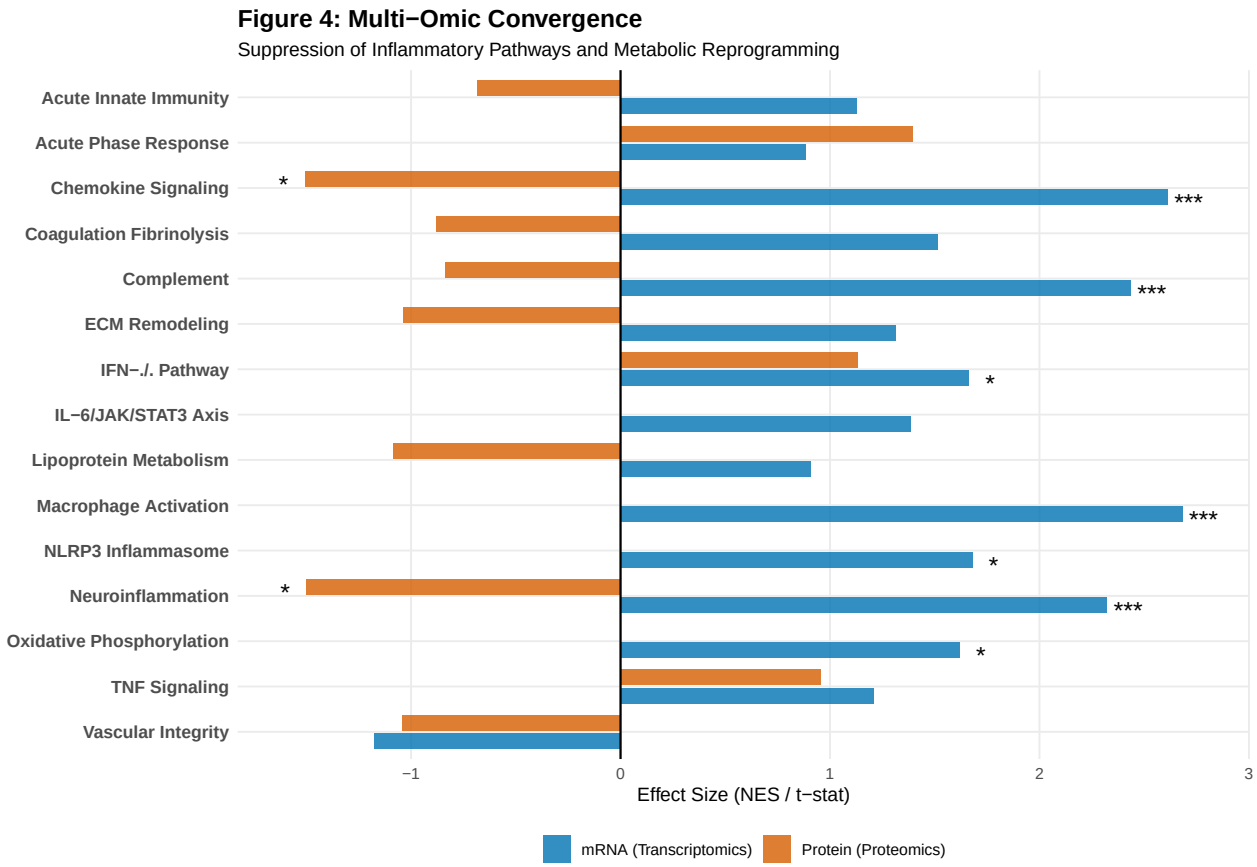
Showing 1 to 10 of 15 entries Previous 1 2 Next

Several findings were consistent across both the proteomics and transcriptomics datasets, revealing a complex regulatory landscape (Table above). Most notably:

- **Transcriptional Enrichment of Macrophage Activation:** Pathways associated with macrophage activation and neuroinflammation showed robust mRNA enrichment ($NES =$

2.68 and 2.32, respectively). This suggests that while cannabis modulates the final inflammatory output, the cells in HIV+ donors remain transcriptionally active or "primed."

- **NLRP3 Inflammasome Trends:** The **NLRP3 Inflammasome** showed significant transcriptional enrichment ($NES = 1.68, p = 0.038$). Given the established protein-level suppression in daily users, this divergence suggests cannabis may act primarily at the level of inflammasome assembly or post-translational regulation.
- **Profound Interaction Specificity:** The **Macrophage Activation** ($p_{int} = 2.59 \times 10^{-6}$) and **Neuroinflammation** ($p_{int} = 1.68 \times 10^{-7}$) pathways demonstrated the most profound specificity. This confirms that the cannabinoid-induced transcriptional "re-tuning" is a targeted response to the HIV+ environment, with minimal impact on baseline HIV-negative MDMs.
- **Metabolic Recovery: Oxidative Phosphorylation** was consistently enriched across platforms ($NES_{mRNA} = 1.62, p = 0.031$), signaling a sustained metabolic shift toward mitochondrial health and bioenergetic recovery in these donors.
-



The "Priming vs. Blockade" Model

A central finding is the significant **HIV-specific modulation** of the **NLRP3 Inflammasome** ($p_{int} = 0.036$). While we observed a robust enrichment of NLRP3-related transcripts ($NES = 1.68, p = 0.038$), this stands in contrast to the established suppression of the inflammasome at the protein level. This divergence suggests that while the

macrophage remains transcriptionally "primed" for an inflammatory response, cannabis exposure may interfere with the **post-translational assembly** or **translational efficiency** of the inflammasome complex.

When viewed alongside the robust enrichment of **Oxidative Phosphorylation** ($NES = 1.62, p = 0.031$), these results indicate a fundamental metabolic and inflammatory "re-tuning." Cannabis appears to facilitate a shift away from the hyper-inflammatory glycolytic state typical of HIV+ MDMs toward a profile of **mitochondrial recovery**, even while maintaining a heightened transcriptional surveillance state.